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BEST U.S. MARKETS FOR HYDROGEN DEPLOYMENT

Regional Opportunity Analysis for On-Site Production Version 2.0 · February 2026 Non-Confidential

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EXECUTIVE SUMMARY

The economic viability of hydrogen as an energy carrier is significantly influenced by regional production and supply chain costs. This document identifies and analyzes key U.S. markets where a compelling arbitrage opportunity exists for on-site hydrogen generation. These markets are characterized by notably high delivered hydrogen prices, often due to their distance from major production hubs, coupled with some of the nation's lowest industrial electricity rates, frequently supported by a substantial share of renewable energy. By examining states such as Washington, New Mexico, South Dakota, Iowa, and Oklahoma, this analysis highlights strategic locations where investment in localized hydrogen production, potentially leveraging green electricity, can offer significant cost advantages over incumbent supply models, thereby accelerating the deployment of hydrogen technologies.

1. INTRODUCTION

As the United States seeks to expand its hydrogen economy, understanding regional market dynamics is crucial for strategic investment and infrastructure development. The delivered cost of hydrogen can vary dramatically across the country, influenced by factors such as proximity to centralized production facilities (primarily on the Gulf Coast and Southern California), transportation logistics, and local demand scales.

Concurrently, industrial electricity prices also exhibit significant regional variation, with some areas benefiting from abundant, low-cost power, increasingly from renewable sources. This analysis identifies U.S. markets presenting a strong economic case for decentralized, on-site hydrogen production. These are

regions where end-users currently face high prices for delivered hydrogen, while simultaneously having access to low-cost industrial electricity.

The presence of a high renewable energy share in the local grid further enhances the potential for producing cost-effective green hydrogen. By ranking these markets, this document aims to pinpoint areas ripe for investment in localized hydrogen generation capacity, which can undercut expensive delivered hydrogen and foster regional hydrogen ecosystems.

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2. RANKED MARKETS FOR ON-SITE HYDROGEN PRODUCTION

This section identifies U.S. markets ranked based on the disparity between high current delivered hydrogen prices and low prevailing industrial electricity rates, with renewable energy penetration noted as a key supporting factor.

2.1 Washington (Pacific Northwest)

- **High Delivered Hydrogen Price, Low-Cost Green Power**: Washington benefits from one of the lowest industrial electricity rates in the U.S., approximately **6.35¢/kWh**, largely due to its extensive hydropower resources.
- **Renewable Energy Profile**: The state's electricity grid is approximately **76% renewable-powered**, predominantly by hydro.
- **Hydrogen Supply Challenge**: Despite this, Washington lacks significant local hydrogen production, forcing industries to rely on imported hydrogen, often via liquid tanker deliveries from distant plants. This results in high delivered hydrogen costs, estimated to be in the range of **$19 per kilogram** in areas distant from supply hubs.
- **Opportunity Assessment**: The combination of cheap, clean electricity and expensive delivered hydrogen creates a strong incentive for on-site green hydrogen production. Tobe Energy's technology can produce hydrogen at ~$1.50/kg using Washington's grid, undercutting delivered prices by **92%**.

2.2 New Mexico (Desert Southwest)

- **Cheapest Industrial Electricity, Costly Hydrogen Imports**: New Mexico boasts the nation's lowest average industrial electricity rates, around **5.75¢/kWh**, driven by inexpensive wind energy and natural gas generation.
- **Renewable Energy Profile**: Renewable sources, primarily wind, contribute approximately **45% of its electricity generation**.

- **Hydrogen Supply Challenge**: Currently, New Mexico has no major merchant hydrogen production facilities; hydrogen must be transported from the Gulf Coast or California. These long-haul logistics contribute to high local hydrogen prices, often approaching **$10/kg**.
- **Opportunity Assessment**: The pronounced disparity between ultra-low electricity costs and high delivered hydrogen prices makes New Mexico a prime candidate for developing new hydrogen production hubs, particularly leveraging its abundant wind resources. Tobe's **$1.81/kg LCOH** target is achievable even without full renewable integration.

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2.3 South Dakota (Upper Great Plains)

- **Wind-Rich Grid, Hydrogen Supply Deficit**: South Dakota features low industrial electricity prices (approximately **8.0¢/kWh**) and an exceptionally green electricity grid, with wind and hydropower accounting for about **56% of in-state generation** — one of the highest renewable shares nationally.
- **Hydrogen Supply Challenge**: Similar to other Upper Plains states, it lacks local hydrogen production capacity. Consequently, industrial hydrogen for applications such as fueling stations or manufacturing must be transported over considerable distances, leading to substantial delivered prices (e.g., heavy-duty vehicle stations in comparable remote regions report costs near **$19 per kg**).
- **Opportunity Assessment**: South Dakota's abundant wind power and high delivered hydrogen costs present a clear opportunity for on-site hydrogen generation. Tobe Energy's technology can leverage existing wind infrastructure to produce green hydrogen at competitive rates.

2.4 Iowa (Midwest Wind Belt)

- **High Wind Penetration, Dependence on Delivered Hydrogen**: Iowa's average industrial electricity rate is approximately **6.9¢/kWh**, significantly below the U.S. average, with its electricity mix being roughly **60% wind-powered** — the highest wind penetration in the country.
- **Hydrogen Supply Challenge**: Despite this, Iowa does not host large merchant hydrogen facilities; most hydrogen is either produced captively (e.g., at fertilizer plants) or imported. Industries requiring bottled or liquid hydrogen (e.g., for electronics manufacturing or fuel cell applications) face premium pricing due to delivery costs, which can approach **$20/kg** in the absence of local supply.
- **Opportunity Assessment**: The state's strong renewable resources and low electricity costs, contrasted with high hydrogen procurement prices, underpin a compelling case for local green hydrogen projects. Iowa's fertilizer industry represents an immediate market for Tobe's technology.

2.5 Oklahoma (South-Central U.S.)

- **Low Electricity Cost, Gaps in Hydrogen Infrastructure**: Oklahoma's industrial electricity price, around **6.3¢/kWh**, is among the lowest in the U.S., supported by inexpensive natural gas and rapidly growing wind generation.

- **Renewable Energy Profile**: Wind energy now constitutes about **35% of Oklahoma's electricity supply**, and the overall renewable share is increasing.
- **Hydrogen Supply Challenge**: Similar to other regions outside the Gulf Coast, Oklahoma is not served by an extensive hydrogen pipeline network and lacks large-scale merchant hydrogen production. Commercial hydrogen is typically delivered from Gulf Coast plants or other distant sources, incurring significant transportation costs and resulting in delivered prices around **$25 per kg** for many customers.
- **Opportunity Assessment**: This disparity indicates a market well-suited for investment in local hydrogen generation, leveraging Oklahoma's wind potential to reduce end-user fuel costs. Tobe Energy's Tulsa facility represents the ideal launch point for commercial deployment.

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3. KEY INSIGHTS AND STRATEGIC IMPLICATIONS

The analysis reveals a consistent pattern: markets situated far from established hydrogen production hubs (e.g., the U.S. Gulf Coast or Southern California) typically experience the highest delivered hydrogen supply costs due to transportation overhead and a lack of local economies of scale.

Many of these same markets, however, benefit from exceptionally low industrial electricity rates, often driven by significant local renewable energy resources such as wind or hydropower. This confluence creates a favorable economic scenario for deploying on-site electrolysis or other decentralized hydrogen production technologies.

Access to cheap electricity allows for more affordable hydrogen generation, directly undercutting the premium prices charged by incumbent suppliers for delivered hydrogen. For instance, while delivered hydrogen in these underserved markets frequently costs approximately **$19/kg**, on-site production utilizing electricity at $0.05–0.06/kWh could yield hydrogen at a significantly lower levelized cost, particularly when factoring in long-term operational benefits and avoided transportation fees.

Regions such as the Pacific Northwest, the Great Plains, and the Desert Southwest therefore rank highly on a combined metric of cost incentive (high delivered H_2 price vs. low electricity price) and clean energy opportunity (high renewable penetration). These areas represent critical focal points for the expansion of the U.S. hydrogen economy, offering conditions where new investments in local hydrogen projects can effectively bridge the current supply-cost gap.

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4. CONCLUSION

The U.S. exhibits significant regional disparities in both delivered hydrogen costs and industrial electricity prices. Markets like Washington, New Mexico, South Dakota, Iowa, and Oklahoma present compelling

opportunities for the deployment of on-site hydrogen production systems. These regions are characterized by high costs for imported hydrogen and access to low-cost electricity, often with a strong renewable energy component.

Investing in local hydrogen generation in these strategically advantageous markets can lead to substantial cost savings for end-users, reduce reliance on long-distance hydrogen transport, and foster the growth of sustainable, regional hydrogen ecosystems powered by increasingly green grids.

Tobe Energy's membrane-free electrolysis technology, with its **>92% system efficiency** (HHV) and **$1.81/kg LCOH** target, is uniquely positioned to capitalize on these regional arbitrage opportunities. The company's planned **Q4 2026** deployment in Tulsa, Oklahoma provides the ideal launchpad for expanding into these high-potential markets.

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